4 Industry Certification Evidence: Short-Term Projects Documentation

Instructor: Mr. Aldrin Torres
School Year: 2026-2027

Course 1: Game Design & Simulation

Project Title: Unity Environment Showcase Project

Objectives:

- Create a themed 3D environment using Unity and free assets.
- Apply lighting techniques and camera composition.
- Design five camera perspectives to showcase the environment.
- Communicate design decisions through a presentation.

Georgia Standards:
GDAS1, GDAS3, GDAS6, GDAS7

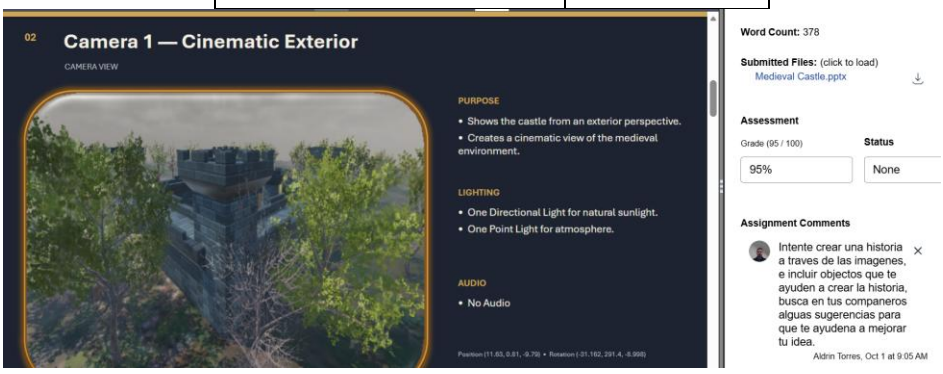
Assignment Sheet

Duration: 5 Class Days / Due October 1st, 2026.
Programming Required: No
Final Submission: PowerPoint and Unity Project.

Students create one Unity scene using a free Asset Store package, at least 10 objects, a clear theme, five camera views, required lighting, and optional audio. The PowerPoint includes a title slide, five camera slides, and a reflection slide.

Minimum Time to Complete: 5 class periods (55 minutes each).

Blank Rubric Criteria	Points
Environment Design	30
Camera Views	30
Lighting	15
PowerPoint Explanations	15
Professionalism	10
Total	100



Course 2: Introduction to Software Technology (IST)

Project Title: Career Exploration Website Project

Objectives:

- Create a professional website using HTML and CSS.
- Research a technology career pathway.
- Analyze certifications, education pathways, and career opportunities.
- Develop professional digital communication skills.

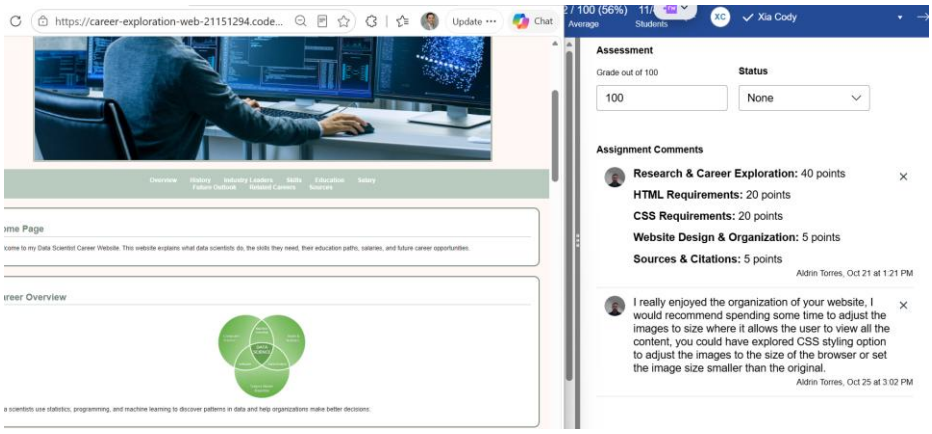
Georgia Standards:
Web development, career exploration, HTML, CSS, and digital product creation standards.

Assignment Sheet

Students create a professional website about a technology career. Required sections include Home Page, Career Overview, Career History, Industry Leaders, Skills Needed, Education & Certifications, Salary Information, Future Outlook, Related Careers, and Sources. HTML requirements include navigation, tables, images, links, headings, and page structure. CSS requirements include styling, layout, hover effects, and consistent design.

Minimum Time to Complete: 4 class periods (55 minutes each).

Blank Rubric Criteria	Points
Research & Career Exploration	40
HTML Requirements	20
CSS Requirements	20
Website Design & Organization	15
Sources & Citations	5
Total	100



Course 3: AP Computer Science Principles (AP CSP)

Project Title: Big Idea 2 Data Technology Research Symposium

Objectives:

- Research a major data technology concept.
- Analyze contributors, inventions, and algorithms.
- Explore industry careers.
- Present findings professionally.

Standards: AP CSP Big Idea 2: Data.

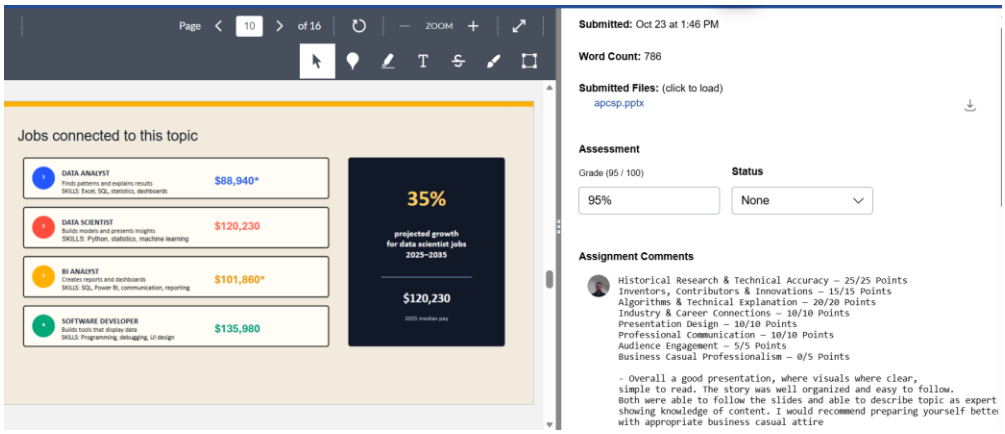
Assignment Sheet

Student teams investigate a data topic including historical background, contributors, inventions, algorithms, industry applications, career pathways, ethics, and future trends. Deliverables include a presentation, research summary, and AP CSP multiple-choice question.

Minimum Time to Complete: 5 class periods (55 minutes each).

Blank Rubric Criteria	Points
Historical Research & Technical Accuracy	25
Inventors, Contributors & Innovations	15
Algorithms & Technical Explanation	20
Industry & Career Connections	10
Presentation Design	10
Professional Communication	10
Audience Engagement	5
Business Casual Professionalism	5
Total	100

Student Work Sample:



Course 4: AP Computer Science A (AP CSA)

Project Title:

Deck of Cards API Game Development Project

Objectives:

- Use a real-world REST API to retrieve live data from the internet.
- Read and process JSON-formatted data returned from an API.
- Create a Java Swing graphical user interface.
- Develop a card-based game using object-oriented programming concepts.
- Demonstrate problem-solving and software development skills through iterative testing and refinement.

AP CSA Standards:

- Unit 2: Selection and Iteration
- Software Development Practices

Assignment Sheet

Project Description:

Students will develop an interactive card game using the **Deck of Cards API**. The provided starter application demonstrates how to connect to an external REST API, retrieve card data in JSON format, extract information from the response, and display card images within a Java Swing GUI.

Students will expand upon the starter code by designing and implementing their own game rules. Possible game options include:

- Highest Card Wins
- Blackjack
- War
- Memory Matching
- Custom Strategy Card Game

Students will use API responses to generate gameplay events and create a complete user experience through Java programming.

Current Course Progress Note:

Implementation Status: This project is scheduled after completion of AP CSA Unit 2 and introductory instruction in file input/output. Students have not yet reached the prerequisite content required for independent completion of the project.

This short-term project is scheduled for later in the semester after students complete **AP CSA Unit 2 (Using Objects)** and receive instruction in:

- File Input/Output
- String Processing
- Collections and Data Management
- Class Design
- Basic Software Architecture

At the time of documentation, students have not yet completed the prerequisite content necessary to successfully implement and extend the project independently. The starter API demonstration is being prepared as the foundation for the upcoming short-term project.

Minimum Time to Complete:

4-5 class periods (55 minutes each)

Final Submission:

- Java Project Source Code
- Game Design Explanation
- Reflection Document

- Screenshot or Recorded Demonstration

Blank Rubric Criteria

Criteria	Points
API Integration & Data Processing	25
Program Functionality	25
Game Design & Creativity	20
Object-Oriented Programming	15
User Interface Design	10
Documentation & Reflection	5
Total	**100**

Industry Certification / Career Readiness Connection

Students gain experience working with technologies commonly used by software developers, including:

- REST APIs
- JSON data processing
- GUI development
- Software debugging
- Application design

The project mirrors real-world software development practices used in web services, fintech applications, gaming systems, and enterprise software. Students also develop technical documentation and problem-solving skills aligned with industry expectations for entry-level software developers.